Generative AI for Systems Thinking: Can a GPT Question-Answering System Turn Text into the Causal Maps Produced by Human Readers?

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Abstract

Representing a system as a network is critical to support systems thinking, hence several tools have been developed to derive networks from text in educational technology, modeling and simulation, or forecasting. Large-Scale Pre-Trained Language Models (PLMs) have recently come to the forefront to create question-answering systems (Q&A) that can extract networks from text. In this paper, we design and implement a Q&A system that uses GPT-3.5 together with 12 filters to extract causal maps text. Our evaluation on two topics via several policy documents finds that GPT accurately extracts relevant concept nodes but occasionally reverses causal directions and struggles with the type of causality as it lacks an understanding of event sequence. We also show that automatically extracted maps can only partially resemble human-made maps collected on the same topics. By making our Q&A system open-source on a permanent repository, researchers can evaluate it with newer PLMs as technology improves.

Keywords: Causal Map, Generative AI, GPT, Network Analysis, Systems Science

1. Introduction

An essential step in systems science is the ability to *represent* a system in a way that supports analytical efforts. The representation depends on the application field and the task of interest (Voinov et al., 2018). For example, a physical system such as the climate can be represented via equations governing atmosphere and ground dynamics, analyzed by a solver to identify trends (e.g., change in precipitation) and affected areas (Sreenath, 1993). In contrast, cognitive and social systems are often represented via *soft systems methodology*, with an emphasis on capturing the mindset of stakeholders, their values and expectations, and the barriers to efficient policy interventions (Hieronymi, 2013). A *network* with more (e.g., causal loop diagram) or less (e.g., rich picture) constraints is often used as representation, for at least two reasons. First, as a *visual* object, a network can be transparently presented to a stakeholder to summarize key elements of an interview, such as the list of factors deemed relevant by the stakeholder and their relationships. For example, a visual depiction of a system can help participants engage with data and discuss its implications (Conte and Davidson, 2020; Giabbanelli and Vesuvala, 2023), while pedagogical initiatives that emphasize systems thinking have often asked learners to create maps as a means to organize concepts regarding a problem domain (Schultz et al., 2022). Second, as a *mathematical* construct, a network can be analyzed to reveal important information about a person's mental model of a system (Blacketer et al., 2021), compare the views of individual stakeholders or groups (Aminpour et al., 2021), identify structural barriers or unintended consequences of interventions (Giabbanelli, Rice, et al., 2022), or reduce the complexity of a large map to its core themes (Hayward et al., 2020).

The aforementioned benefits can only be unlocked once a network representation has been obtained. There has thus been considerable interest in automatically *deriving networks from data*. Textual data has attracted significant attention as a data source, for two different applications (Fig. 1). First, several tools have been developed for over a decade in educational technology (Aguiar et al., 2016; Burkhart et al., 2020; Qasim et al., 2013; Villalon and Calvo,

2011; Yang et al., 2022) so that educators could transform the written essays of their students into maps and then perform assessments through a dedicated network analysis software (Wang and Giabbanelli, 2023). Similar works have focused on extracting a map from an expert's text and using network analysis to dissect key concepts that should be passed onto students (Kim et al., 2020). Second, in a separate strand of literature, simulation and forecasting experts turn texts into maps for conceptual modeling and scenario-based foresight, respectively. Texts would be obtained (e.g., automatically via a web crawler) from sources that provide rich descriptions of the structure and dynamics of systems, such as expert interviews, white papers, or research articles (Bishop et al., 2007).

A recent review showed that artificial neural networks are quickly gaining traction as a digital technology to convert text into such maps (Ködding et al., 2023). In particular, deep neural networks such as Large-Scale Pre-Trained Language Models (PLMs) can be given a source text and guided through a conversation to gradually tease out a map, as simplified in Fig. 2. Such prototypes have been built on BERT and produced either a mind map (Feblowitz et al., 2021) or a form of causal map (Davis et al., 2022). This specific use of a PLM is known as a *question-answering* (Q&A) system. Starting with the text and an initial topic or question, the Q&A system retrieves the answer from the text. Parsing the answer can generate additional questions, resulting in a 'conversation' about the text input. In Fig. 2, questions are about causes and consequences, and the answers are parsed to isolate concepts.

While such systems are increasingly familiar in e-commerce platforms (Du et al., 2023), there is also a growing interest in using them to identify relations (Yan et al., 2021). The introduction of accessible and disruptive technology such as generative AI has allowed to create GPT-based Q&A systems (Omar et al., 2023). The literature reports mixed findings, as some authors noted that GPT-3.5 outperforms well-known models such as ChatGPT and T5 on causal questions (Tan et al., 2023), while others observed that GPT-3.5 models only "have outstanding performance and robustness for some specific relational judgments" found in common expressions (Chen et al., 2023). In addition, there are ongoing debates in research on generative AI about how we *evaluate* the accuracy of such systems. They may be incorrect when identifying concepts, as *hallucination* can cause the answer to be a reflection of the knowledge model derived from training data rather than the text input. They could also report relations that do not exist, or with the wrong type of causation. Automatic (often based on string-matching) and human evaluations (e.g. via crowdworkers) have shortcomings (Xu et al., 2023) to assess factuality.

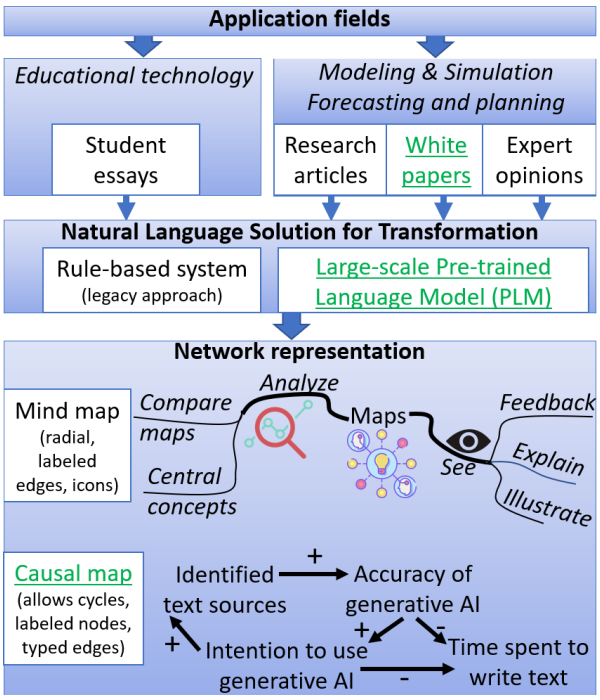

Figure 1: Deriving a network from text is an important task in several application domains, with several tools originating from research communities including educational technology, modeling & simulation, or forecasting. Once a network is obtained, it can be used as a visual aid or analyzed. To situate this paper within the literature, our techniques and data sources are underlined and colored in green.

In this paper, we propose an approach to derive causal maps from text using GPT 3.5 as a PLM and we evaluate it on white papers by contrast to human-generated causal maps on the same topics. Our main contributions are twofold:

① We perform prompt-engineering to guide GPT in identifying causal relationships and we empirically devise a set of 12 filters to extract results in a conservative manner that limits the risk of false-positives. This contributes to the nascent literature on using a PLM to identify causal relationships.

② We compare the characteristics of the map produced by our approach with maps created by individuals from the same large source documents (40+ pages). We use documents from two domains (obesity and suicide) and employ several individuals trained in systems thinking. As human maps vary, to derive *distributions* of their characteristics and position the GPT-solution with respect to these distributions. Our approach

to evaluation goes beyond classic string-matching techniques and constitutes a hybrid between human evaluations (since they create maps) and automatic evaluations (which derive characteristics from maps).

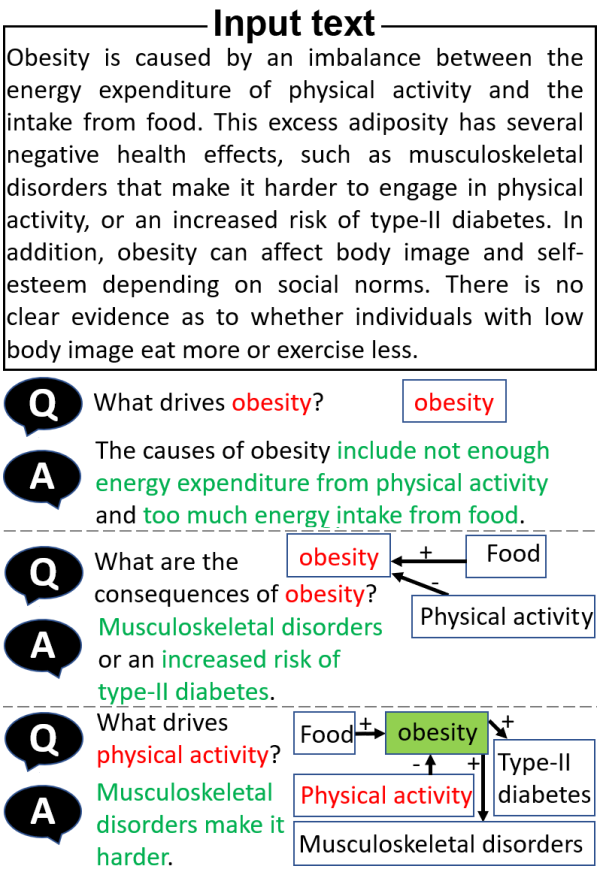

Figure 2: The Q&A system is given a text and starting topic ('obesity'). It starts by automatically asking the causes and consequences of obesity. Results are parsed to extract topics and the type of causation. Once obesity is covered, the system follows-up on each topic at distance 1 (starting with 'physical activity'), then at distance 2, and so on. The maximum distance can be user-defined.

The remainder of this paper is organized as follows. We provide a succinct background on the analysis of maps in section 2 to keep the paper self-contained with respect to the structural characteristics that are used in our evaluation. Then, we present our approach in section 3 with an emphasis on filters, and we evaluate it in section 4 on two domains and four documents. Finally, we discuss these results in section 5 with respect to the feasibility of using GPT-3.5 models to extract causal maps from text. To support transparency and replicability efforts, our scripts are permanently available at https://doi.org/10.5281/zenodo.8045304.

2. Background: Analyzing Maps

The analysis of a systems map borrows from graph theory, hence the term *graph* is used instead of map in this section. A graph can be characterized at several levels. At the *graph-level*, a metric would return a single number, such as the number of nodes. At the *element-level*, we produce one number per element (e.g., the number of edges for each node, known as *degree*) and we either study distributions or focus on the top elements (e.g., most central nodes). As an intermediate, an analysis can be conducted at the *group-level*, such as the number of components. Maps of cognitive and social systems are normally connected hence the number of components is expected to be 1. They are small- to medium-sized hence we do not expect significant variations in distributions at the element-level, unlike e.g. degree distributions in large-scale networks. The most important nodes are often a reflection of the elicitation process, e.g. if the conversation starts with 'obesity' then it is likely that obesity will be at the center of the system. Consequently, analyses for causal maps and related representations are predominantly conducted at the graph-level, as illustrated in studies such as (Shahvi et al., 2021; Wang and Giabbanelli, 2023). In this section, we provide an intuitive and succinct summary of graph-level metrics. For a comprehensive overview along with equations, we refer the reader to Sec. 3.2 'Structural Analysis' in (Tchupo and Macht, 2022).

The *size* of the graph is characterized with respect to the *number of nodes* and *number of edges*. The *average degree* measures the mean number of edges per node. For the causal map in Fig. 1, this would result in an average degree of $\frac{2+3+2+3}{4} = 2.5$; note that it is not equivalent to simply summing up the number of edges and dividing by the number of nodes (5/4 = 1.25). The *density* measures the number of existing edges in the map as a ratio of the maximum number of edges that the map could have if every node was linked to every other node. Given the same example, the density would be $\frac{5}{4\times3} = 0.416$. Unlike mind maps which are rooted trees and hence forbid cycles, causal maps frequently contain cycles. The *average cycle length* thus measures the mean length (i.e., number of edges) of the cycles. When there is a single cycle (as in Fig. 1), the average is the same as the length of this cycle (3). Finally, the *average distance* measures the mean smallest number of edges required to go between every pair of nodes.

The measurements should be contextualized given the expected values for causal maps (Giabbanelli, Rice, et al., 2022). These maps are usually small- to mid-sized, under 100 nodes. They are *sparse*, because

individuals are selective when they create the maps hence they do not see each node as connected to most other nodes. Consequently, the density tends to increase as the number of nodes increases. Finally, because the maps are often created in a 'snowball' approach that focuses on a problem of interest and gradually teases out layers of causes and consequences, the overall average distance in the map is small.

3. Methods

An overview of our methods is provided in Fig. 3. We start with the data, consisting of a white paper as a PDF, which our system will turn into a causal map. Each white paper also contains references, which need to be deleted to prevent our Q&A system from mixing the titles of works cited with the evidence-base summarized in the document. We thus automatically search for a reference Sec. header and remove all content therein. This produces a text file, which we ultimately seek to pass onto GPT along with questions. GPT performs a natural language understanding task known as *Open Domain Question Answering* (ODQA) (Siriwardhana et al., 2023). It receives a question, then selects relevant parts in the text and uses them to generate an answer. We thus have to craft questions carefully (known as *prompt engineering* – see Lo, 2023) and parse elements of the answer, which can generate additional queries for clarification. These two aspects are detailed in dedicated subsections below.

3.1. Finding causal statements

When conducting an interview to create a causal map, the interviewer would start with a problem of interest such as obesity or suicide attempt. In a similar way, the system needs to anchor the causal map in a list of seed terms defined by the user (e.g., 'obesity' and 'overweight'). Given these seed terms, our intent is to pass the *entire document*, query GPT about causes, parse the effects to ask about further causes, and so on. However, the GPT API has a token limit[1], so a large file cannot fit within one prompt. Consequently, we send several prompts, each one aggregating as many complete sentences as can fit (with a safety margin) within a prompt. Each prompt uses the following template, where 'factor' could be replaced by 'obesity':

[1] A prompt has a limit of 4,096 tokens, but a token is not equivalent to one word. A token covers approximately 4 characters in English (OpenAI, 2023b), and the number of words and characters vary across sentences.

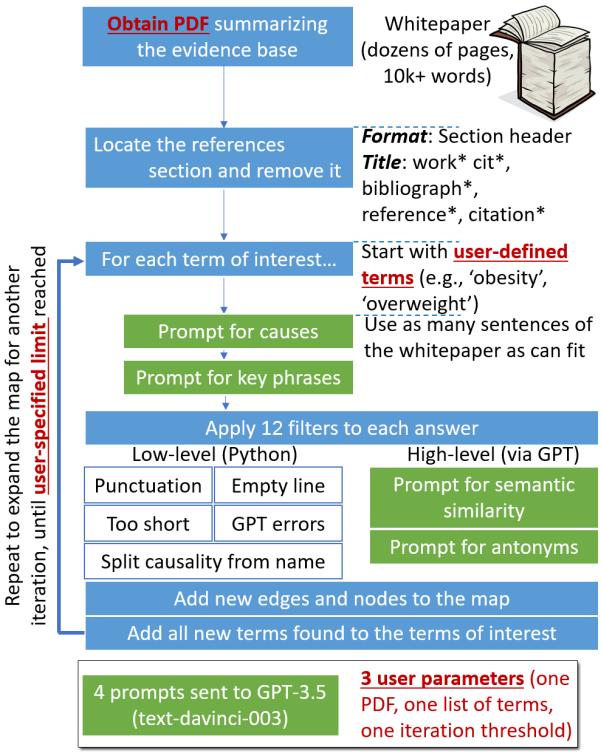

Figure 3: Our system requires three **user inputs** and sends several **prompts** to GPT. The first two prompts are explained in Sec. 3.1 and the later two are explained alongside other filters in Sec. 3.2.

> **[Question prompt.]** With the text below answer the questions. Insert aggregated sentences from text here. In this text, what causes factor to increase?
> If there are no answers in the text, do not answer anything. The answer should be as short as possible.

A sample result from GPT could be as follows:

> **[GPT sample answer.]** The cause of the increasing rate of obesity is believed to be due to a combination of lifestyle, exercise, diet and nutrition. Additionally, economic factors, such as the cost of healthcare for overweight and obese children, may contribute to the obesity epidemic.

The result needs to be divided into a list of concepts, hence we follow up with a second short prompt:

[Question prompt.] Extract each keyphrase from this text: insert answer to previous prompt here. Give me each keyphrase separate by a \n.

Conceptually, these two prompts could be combined. However, we observed through experiments with GPT that directly asking for each keyphrase from the source document often produced only one keyphrase whereas more results were generated when asking for keyphrases within sentences selected for causality by GPT. In the example above, a sample answer could be ["combination of lifestyle, exercise, diet and nutrition.", "economic factors", "cost of healthcare", "obesity"].

3.2. Filtering answers

As shown above, GPT's answer does not directly produce a list of causal elements hence we employ a series of 12 filters to extract them (Table 3). The list of filters was inspired by the design of earlier prototypes (Davis et al., 2022), which themselves often inherit practices from natural language processing for social media analytics. The specific filters and their order were determined through trial-and-error. Filters belong to two broad categories. *Low-level filters* involve a few lines of Python code and are guaranteed to work. They support operations such as separating elements of the answer (e.g., to separate exercise and diet), removing punctuations (so that 'nutrition.' and 'nutrition' can map to the same construct), and eliminating obvious redundancy (e.g., obesity is a keyphrase in the text but not a cause for obesity itself).

High-level filters involve advanced Natural Language Understanding and send additional prompts to GPT, hence results vary. These filters identify constructs in the answer that are equivalent to existing nodes in the map, to avoid including redundant nodes in the map (Fig. 4a). One form of equivalence is *semantic similarity* (Fig. 4b), which is based on closed related meanings (e.g., 'impulsivity' and 'acting suddenly without planning of considering the consequences'). Identifying semantically similar terms has long been a challenge (Chandrasekaran and Mago, 2021), particularly for causal mapping where approaches have included word matching, recommender systems, or thesauri (Giabbanelli and Tawfik, 2019). PLMs have been successfully used for semantic similarity detection (Peinelt et al., 2020), including in the context of Q&A systems (Al-Besher et al., 2022), hence we leverage this recent advance in our system. Antonyms form the other equivalence (i.e., words with opposite meanings such as 'unhealthy eating' and 'healthy eating'). We detect them with GPT and integrate them in the map by reversing causal connections (Fig. 4c).

Prompt engineering was necessary to minimize the number of errors made by GPT when identifying equivalent terms. We only created prompts with terms that would not be relevant to obesity or suicide, to avoid affecting our evaluation based on these two domains in section 4. For semantic similarity, we created four versions of the prompt based on whether to provide examples (on/off) and counter-examples (on/off). We measured the number of errors in several examples (e.g., Table 2) and settled on the prompt below as it minimizes the number of errors:

[Question prompt.] Extract all keyphrases that are semantically similar in this list: insert list of candidate terms here and display them as a tuple. \n You should only use words with the same semantic meaning and not words with opposite meanings. For example, in the following list ['work hard', 'work very little', 'arrive on time', 'money','put in a lot of effort', 'punctuality'], the result should be [(work hard, put in a lot of effort), (arrive on time, punctuality), (money, income)]. \n Each key phrase must be separated so that the result is a list of key phrase pairs. Note that the key phrases must be in lower case and must not contain any punctuation.

The same process was performed for antonyms, arriving at the following prompt:

[Question prompt.] Extract all antonyms of insert term here in this list insert list of candidate terms here and display them as a list. You should only take words with opposite meanings. For example, in the following list ['work hard', 'work very little', 'arrive on time', 'money','put in a lot of effort', 'punctuality'], the result for the word 'unpunctuality' should be ['arrive on time', 'punctuality']. Note that the key phrases must be in lower case and must not contain any punctuation.

4. Results

Our experiments were performed on three datasets (Table 1) using maps available on open repositories, as described in (Giabbanelli, Rice, et al., 2022) and (Freund and Giabbanelli, 2021) for suicide and obesity, respectively. Instead of a model optimized

Table 1: We use two white papers on obesity and one on suicide. Two of the datasets have several human-made maps (which allows to position our GPT-generated map within a distribution) while another has a single map.

Dataset	# of human maps	Types of participants	Documents	
			PDF Pages	Words
Suicide	19	Subject-matter experts with trained facilitation	108	21,982
Obesity- 2014P	1	Trained students reading a policy document from Pennsylvania (P) or the Institute of Medicine (IOM)	77	24,578
Obesity- IOM	5		46	10,330

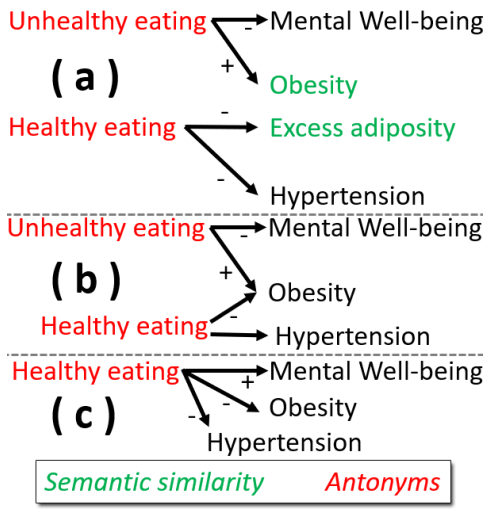

Figure 4: Constructs extracted from GPT's keyphrases may be redundant (a). We identify pairs of **semantically similar nodes**, remove a node and attribute all edges to the remaining node (b). We also identify **antonyms**, remove a node and reverse the causal direction of its edges when attributing them to the remaining node (c).

for code-completion (code-davinci-002) or chat (gpt-3.5-turbo), we used the GPT-3.5 davinci-003 that is positioned for "any language task with better quality, longer output, and consistent instruction-following than the curie, babbage, or ada models" (OpenAI, 2023a).

To understand the outputs and identify patterns of errors, an expert[2] investigated the content of maps from GPT at three increasingly fine-grained levels of systems thinking: whether nodes were clear and relevant, whether the endpoints and directions of edges were correct, and whether the right causality (increasing/decreasing) was assigned. First, we found that the system is generally effective at distilling large white papers into a smaller list of relevant constructs (Table 5). The system is able to solve elaborate cases of semantic similarity but struggles with obvious ones (e.g., 'child obesity' and 'childhood obesity', 'healthful food' and 'healthful foods'), which caused all the mistakes for the suicide case and several mistakes for obesity[3]. We also noted that several constructs cannot be interpreted without context (e.g., 'lack of evidence', 'shift in focus'). Second, accuracy at the level of edges was generally satisfactory, and most problems would be solved by reversing the edges (i.e., $A \rightarrow B$ should be $B \rightarrow A$). As a note of concern in the suicide case, GPT views that suicide solves financial problems and chronic pain. The white paper most certainly did not endorse such views, as it is the prevention package from the Centers for Disease Control and Prevention (CDC). Third, the type of causation is poorly understood across applications, as the system was widely inconsistent (e.g., public health campaigns *increased* overweight but *decreased* obesity). An interesting case is the suicide map: GPT only used risk factors (although the document contains an abundance of protective factors), hence every causation should have been increasing. However, 45% of edges were mislabeled as a decrease. Common issues were a lack of understanding of the temporal sequence of events. For example, GPT views that stress *will* increase *prior* suicide attempts, and controversial

Table 2: We provided 8 candidate terms: overweight, obesity, physical inactivity, unhealthy eating habits, healthy food options, cardiovascular disease, lack of physical activity, fat. Expected pairs were (obesity, fat) and (physical inactivity, lack of physical activity). If a term was not found in an expected pair, we counted it as one error. We counted a wrong pair such as (unhealthy eating habits, healthy food options) as two errors.

Examples	Counter-examples	Errors
Off	Off	8
On	Off	1
Off	On	2
On	On	3

[2]Faculty member with at least 3 years of experience in leading and publishing research relevant to the application domain.

[3]This is a design flaw of the filters rather than a reflection of GPT's capabilities. The 12 filters in Table 3 parse each answer *independently*, hence they only detect antonyms/synonyms within one answer. If GPT uses a term in one answer and a synonym in another, this similarity would be missed. Newer versions of our system have created a dictionary as a supplemental filter to remedy this shortcoming.

Table 3: We use 12 filters, divided into low-level Python filters (1–9 and 12) and high-level filters (10, 11) that send additional prompts to GPT. Our guiding example illustrates that GPT returns part of the prompt (here 'keyphrase') when it fails to answer. Large key phrases must be divided (e.g., [3]). The last filter can eliminate a large number of results that are correct for the application field, but are too anecdotal to be included in a causal map.

Filter	Action	Guiding example
1	Lower capitalization	[1] keyphrase,demographic shift of population ,↓,-1 [2] "processed foods, increase",processed foods , ↑,1 [3] "high-energy-dense foods, processed foods, fat and/or sugar, low in nutrients, portion sizes, restaurants, grocery stores, vending machines.",high-energy-dense foods ,↓,-1
2	Remove lines that are either empty or equal to the prompt (indicates a GPT error)	Line [1] is removed as 'keyphrase' was from the prompt
3	Split the list of factors	processed foods,processed foods , ↑,1 increase,processed foods ,↓,-1 high-energy-dense foods,high-energy-dense foods ,↓,-1 processed foods,high-energy-dense foods ,↓,-1 ... grocery stores,high-energy-dense foods ,↓,-1 vending machines,high-energy-dense foods ,↓,-1
4	Remove final period	processed foods,processed foods , ↑,1 high-energy-dense foods,high-energy-dense foods ,↓,-1 processed foods,high-energy-dense foods ,↓,-1 ... grocery stores,high-energy-dense foods ,↓,-1 vending machines,high-energy-dense foods ,↓,-1
5	Remove sentences >5 words (whole sentence instead of keyphrase) or <4 characters (not a full word)	
6	Remove punctuation signs	
7	Remove extraneous white spaces	
8	Remove words starting with increas* or decreas* and updated causal connections	
9	Remove cases where causes = effect	processed foods,high-energy-dense foods ,↑,1 fat and/or sugar,high-energy-dense foods ,↑,-1 ... vending machines,high-energy-dense foods ,↓,-1
10	Combine based on semantic similarity	
11	Modify based on antonyms	
Re-apply filters 1 to 7		
12	Drop causal terms that appeared only once	processed foods ,high-energy-dense foods, ↑,1

views surrounding death by suicide may also be ascribed to ignoring the finality of death.

Using the six graph-level characteristics of maps explained in section 2, we compared the maps produced by our system at several levels of iteration with the maps produced by humans (Table 4). As expected, the number of iterations is a key parameter: it determines the size of the map and hence affects other metrics. Overall, the system can create maps that closely resemble human maps in *some*, but not all, characteristics. Density can often be matched, hence the system is parsimonious in identifying causal statements, just as humans are selective when mapping a problem domain. The number of nodes can be matched *if the document is long*, but the number of edges tends to be too high. While the automatic maps cannot always provide a close match, their characteristics are often within the distributions inferred from human-made maps (Fig. 5).

We emphasize that this comparison only seeks to identify the *differences* between automatic maps and their human counterparts. This is a different matter from *correctness*, as human-made maps can also be fallible. For example, the trained human reader for Obesity–2014P failed to find a single cycle, whereas the GPT-based system identified several.

5. Discussion

In this paper, we examined the feasibility of using GPT-3.5 to extract causal maps from text. Our system has three parameters, two of which are easily interpretable (terms defining the topic of interest and document to parse) and only one needs to be adjusted by the user (number of iterations). By detailing the design of our system and providing its code on an open repository, our work can be used with a

Table 4: Characteristics of the maps produced by our solution at different number of iterations, compared to the maps either directly created by human readers for obesity or created by a facilitator from interviews about suicide.

Dataset	Metric	Human map				GPT map by iterations		
		Average	Std	Min	Max	1	2	3
Suicide	#nodes	61.19	20.06	29	100	16	40	188
	#edges	83	28.12	38	126	26	107	851
	avg. degree	1.36	0.21	1.05	1.94	1.63	2.68	4.53
	density	0.05	0.02	0.02	0.09	0.22	0.14	0.05
	avg. cycle length	4.96	1.23	3.33	7.56	4.36	4.13	4.39
	avg. distance	3.72	0.77	2.73	5.91	1.89	2.26	2.35
Obesity -2014 P	#nodes	62				14	60	204
	#edges	54				24	131	652
	avg. degree	0.87				1.71	2.18	3.20
	density	0.03				0.26	0.07	0.03
	avg. cycle length	0				4	5.63	5.45
	avg. distance	2.63				1.74	2.73	2.85
Obesity -IOM	#nodes	26.25	8.77	15	36	3	6	9
	#edges	40	12.11	32	58	2	7	9
	avg. degree	1.63	0.55	0.94	2.13	0.67	1.17	1
	density	0.16	0.11	0.05	0.3	0.67	0.47	0.25
	avg. cycle length	3.49	0.26	3.11	3.67	0	3	3
	avg. distance	2.22	0.36	1.84	2.70	1.33	1.6	1.86

Table 5: Manual evaluation for the accuracy of maps created at two iterations.

Dataset	Nodes	Edges	Causation
Suicide	75%	80%	55%
Obesity-2014P	77%	100%†	82%†
Obesity-IOM	100%	87%	28%

†The few erroneous nodes were involved in many edges. We did not assess their direction or causation, to avoid double counting the same mistake.

different PLM such as LLaMA-based models (e.g., Alpaca, Dromedary), which are smaller and may provide comparable or potentially better performances on some datasets (Sun et al., 2023). When using GPT-3.5, we found it possible to effectively synthesize large white papers into a small set of relevant constructs for a problem domain. However, GPT-3.5 struggles with the direction and nature of causation, hence we recommend limiting these maps to associations (i.e., undirected graphs). We also note that fine-tuning the iteration parameter can create maps whose structure *partially* resembles human-made maps, as it is possible to match some aspects (e.g., number of nodes or density) but not all. Our inspection of errors revealed that GPT-3.5 did not have notable issues of hallucination and excelled at finding synonyms/antonyms, but the lack of understanding regarding the sequence of events produced erroneous and occasionally controversial causal connections. Our system could thus serve to compare complex systems depicted in white papers by turning them into undirected graphs and leveraging the abundant literature on graph comparison.

There are two interesting avenues for future research. First, facilitators build maps with human participants by asking follow-up questions until an answer becomes tangential. For example, frequent household quarrels are a risk for suicide ideation in children, and a divorce can fuel such tensions. However, a facilitator would not venture into detailing the risk factors for divorce when building a map of youth suicide. Based on this observation, the system could *locally* adapt the number of iterations until hitting an out-of-topic concept, instead of being constrained by a global parameter. Second, many PLMs such as GPT rely on their billions of trained parameters to encode their knowledge of the world. However, systems such as *Retrieval Augmented Models* (RAGs) can use both parametric-memory (to leverage their own knowledge like GPT) and non-parametric memory (to extract data from external knowledge bases). Such design benefits various forms of question-answering systems by reducing hallucinations (Hu et al., 2023). Our system could thus be evaluated on RAGs by using the evidence-base from white papers and other official sources as external knowledge bases.

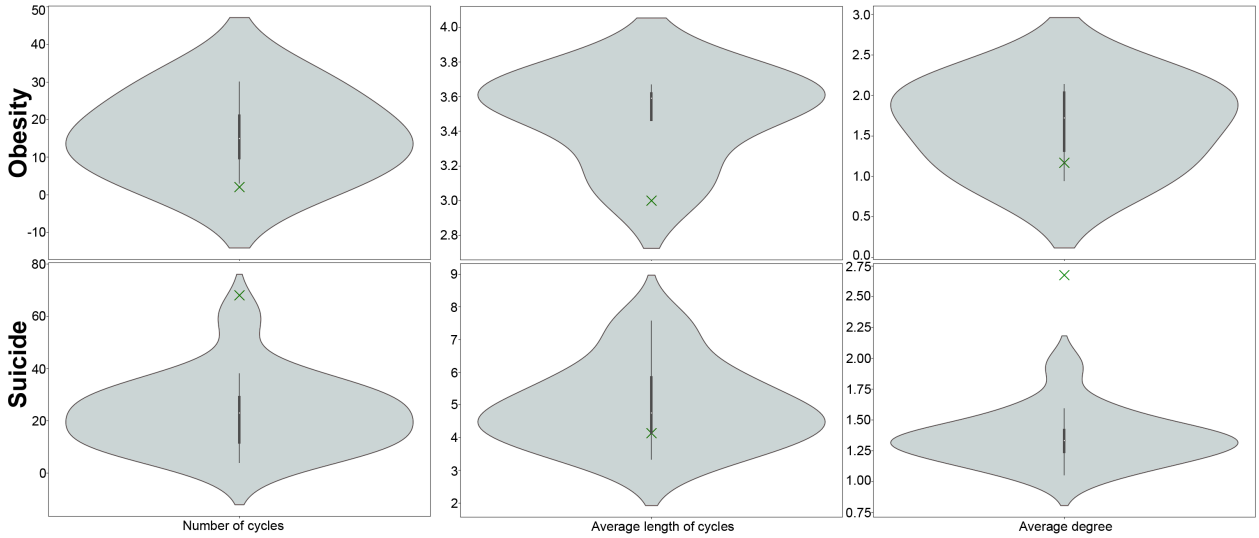

Figure 5: Distribution of characteristics for human-made maps (violin plots) compared to the map automatically derived by the GPT-based system (marked with $\times$) after two iterations. Our repository includes all distributions.

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